EXPERIMENT 26: ANTICONVULSANT ACTIONS OF DRUGS

40 STANDARD IN-COURSE MCQs

1. The primary aim of Experiment 26 is to evaluate:

A. The analgesic properties of NSAIDs B. The anticonvulsant actions and protective efficacy of drugs in mice C. The local anesthetic potency of procaine D. The sedative duration of barbiturates alone

Answer: B

2. What experimental animals are designated in the protocol on Page 133?

A. Adult female guinea pigs B. Adult male albino mice C. Adult female Wistar rats D. Adult male New Zealand rabbits

Answer: B

3. Why are adult male mice specified rather than female mice in this assay?

A. Female mice never develop convulsions B. Estrous cycle hormonal fluctuations can alter seizure threshold and drug metabolism C. Male mice do not have glycine receptors D. Female mice are completely resistant to picrotoxin

Answer: B

4. Which standard barbiturate anticonvulsant is utilized in this experiment?

A. Thiopentone sodium B. Phenobarbitone sodium C. Hexobarbitone sodium D. Amobarbitone sodium

Answer: B

5. What dose and route are specified for phenobarbitone administration in Group I?

A. 5 mg/kg oral B. 20 mg/kg intraperitoneal (i.p.) C. 50 mg/kg subcutaneous D. 100 mg/kg intramuscular

Answer: B

6. Which benzodiazepine anticonvulsant is tested in Groups III and IV?

A. Lorazepam B. Diazepam (2 mg/mL) C. Midazolam D. Clonazepam

Answer: B

7. What dose and route are used for diazepam in Group III?

A. 0.5 mg/kg oral B. 2 mg/kg intraperitoneal (i.p.) C. 10 mg/kg subcutaneous D. 20 mg/kg intramuscular

Answer: B

8. Which chemically induced seizure model is utilized in Groups I and II?

A. Maximal electroshock (MES) B. Picrotoxin-induced seizures (2 mg/kg i.p.) C. Strychnine-induced seizures D. Pentylenetetrazol (PTZ) infusion

Answer: B

9. Which chemical convulsant is utilized in Groups III and IV?

A. Picrotoxin B. Strychnine (1 mg/kg i.m.) C. Isoniazid D. Bicuculline

Answer: B

10. What is the route of administration for strychnine in Groups III and IV?

A. Oral (p.o.) B. Intramuscular (i.m.) C. Intraperitoneal (i.p.) D. Subcutaneous (s.c.)

Answer: B

11. What is the molecular mechanism of action of picrotoxin?

A. Competitive antagonist at glycine receptors B. Non-competitive blocker of GABA_A receptor chloride channels C. NMDA glutamate receptor agonist D. Voltage-gated sodium channel opener

Answer: B

12. What is the molecular mechanism of action of strychnine?

A. GABA_A receptor agonist B. Potent competitive antagonist at inhibitory glycine receptors in the spinal cord C. Voltage-gated potassium channel blocker D. Acetylcholinesterase inhibitor

Answer: B

13. How does phenobarbitone exert its anticonvulsant effect at the GABA_A receptor?

A. Increases the frequency of chloride channel opening B. Increases the duration of chloride channel opening C. Directly blocks glutamate synthesis D. Inhibits glycine reuptake

Answer: B

14. How does diazepam exert its anticonvulsant effect at the GABA_A receptor?

A. Increases the duration of chloride channel opening B. Increases the frequency of chloride channel opening C. Blocks calcium influx through T-type channels D. Inhibits voltage-gated potassium channels

Answer: B

15. What are the tested oral doses of the unknown plant extracts (A and B)?

A. 5, 10, 15, 20 mg/kg B. 50, 100, 200, 400 mg/kg p.o. C. 25, 50, 75, 100 mg/kg D. 500, 1000, 1500, 2000 mg/kg

Answer: B

16. What stock concentration is prepared for unknown drugs A and B?

A. 10 mg/mL B. 100 mg/mL C. 2 mg/mL D. 500 mg/mL

Answer: B

17. What pre-treatment time is observed between administering test drugs and convulsants?

A. 5 minutes B. 15 minutes C. 30 minutes D. 2 hours

Answer: C

18. What vehicle and dose are administered to control mice in Group I?

A. Normal saline (10 mL/kg) B. Distilled water (2.5 mL/kg i.p.) C. Liquid paraffin (1 mL/kg) D. Olive oil (5 mL/kg)

Answer: B

19. What specific device is used to administer unknown drugs A and B orally?

A. 25G hypodermic needle B. Oral feeding cannula (gavage needle) C. Tuberculin syringe without needle D. Glass dropper

Answer: B

20. According to the protocol, what is the defined criterion for protection?

A. Any mouse that sleeps for more than 1 hour B. Any mouse that does not convulse within 30 minutes after convulsant administration C. Any mouse that shows increased grooming D. Any mouse that does not gain weight

Answer: B

21. What behavioral parameters must be recorded following convulsant injection?

A. Food intake and water consumption B. Time of onset, duration of convulsions/behavioral changes, and death/protection C. Urine output and core body temperature D. Pupil diameter and heart rate

Answer: B

22. Strychnine-induced convulsions are typically characterized by:

A. Absence seizures with 3 Hz spike-and-wave discharges B. Generalized tonic-clonic convulsions with powerful opisthotonos and extensor spasms C. Isolated tail flicking only D. Immediate flaccid paralysis

Answer: B

23. Why does strychnine cause massive extensor thrusts in animals?

A. Direct stimulation of skeletal muscle actin-myosin crossbridges B. Selective blockade of Renshaw cell recurrent inhibition in the spinal cord C. Activation of beta-adrenergic receptors D. Release of acetylcholine at the neuromuscular junction

Answer: B

24. Picrotoxin convulsions primarily originate from:

A. Disinhibition in the forebrain and cerebral cortex via GABA_A blockade B. Neuromuscular junction stimulation C. Peripheral sensory nerve endings D. Adrenal medullary adrenaline secretion

Answer: A

25. A mouse weighs 0.025 kg and requires 20 mg/kg phenobarbitone from a 10 mg/mL solution. The required volume is:

A. 0.01 mL B. 0.05 mL C. 0.10 mL D. 0.50 mL

Answer: B

26. A mouse weighs 0.030 kg and receives unknown A at 200 mg/kg from a 100 mg/mL stock. The required volume is:

A. 0.02 mL B. 0.06 mL C. 0.12 mL D. 0.60 mL

Answer: B

27. If 4 out of 5 mice in a treated group do not convulse within 30 minutes, the percentage protection is:

A. 20% B. 50% C. 80% D. 100%

Answer: C

28. In Question 3 on Page 133, a "seizure" is scientifically defined as:

A. A popular non-clinical synonym for fainting B. A paroxysmal, excessive, hypersynchronous electrical discharge of cerebral cortical neurons C. Any involuntary contraction of gastrointestinal smooth muscle D. An acute allergic reaction mediated by IgE

Answer: B

29. The term "convulsion" specifically refers to:

A. The electrical discharge in the brain only B. The violent, involuntary motor contractions of skeletal muscles during a seizure C. A chronic genetic tendency to lose consciousness D. A brief psychic absence without motor activity

Answer: B

30. The clinical diagnosis of "epilepsy" requires:

A. A single isolated febrile seizure in childhood B. A chronic neurological disorder characterized by two or more unprovoked, recurrent seizures C. Severe muscle stiffness following intense exercise D. Toxicity caused by overdose of strychnine

Answer: B

31. Non-technical or lay people often use which word to refer to sudden uncoordinated episodes?

A. Absence B. Fits C. Status epilepticus D. Chorea

Answer: B

32. Which convulsant model is standard for screening drugs effective against human absence (petit mal) seizures?

A. Strychnine model B. Pentylenetetrazol (PTZ) / GABA antagonism model C. Electroconvulsive shock extensor model D. Acetic acid writhing model

Answer: B

33. Which drug is considered the clinical drug of choice for terminating status epilepticus acutely?

A. Oral phenobarbitone B. Intravenous diazepam or lorazepam C. Ethosuximide D. Carbamazepine

Answer: B

34. A major adverse effect of phenobarbitone that limits its everyday clinical use is:

A. Marked central nervous system sedation and cognitive slowing B. Acute hypertension C. Severe diarrhea D. Muscle hypertrophy

Answer: A

35. Which hepatic enzyme effect is associated with chronic phenobarbitone administration?

A. Irreversible enzyme inhibition B. Potent induction of hepatic cytochrome P450 enzymes C. Complete destruction of hepatocytes D. Inability to metabolize carbohydrates

Answer: B

36. Diazepam belongs to which pharmacological class?

A. Barbiturates B. Benzodiazepines C. Hydantoins D. Iminostilbenes

Answer: B

37. If an unknown plant extract protects against picrotoxin but NOT strychnine, its mechanism likely involves:

A. Blockade of glycine receptors B. Enhancement of GABAergic neurotransmission C. Activation of sodium channels D. Inhibition of acetylcholine release

Answer: B

38. If an unknown plant extract protects against strychnine seizures, it most likely modulates:

A. Glycinergic or spinal inhibitory pathways B. Gastric proton pumps C. Peripheral muscarinic receptors D. Alpha-adrenergic receptors

Answer: A

39. In comparing efficacy and potency (Question 2 on Page 133), "efficacy" refers to:

A. The dose needed to produce 50% of the maximum effect B. The maximum anticonvulsant protection achievable regardless of dose C. The speed of absorption from the gut D. The molecular weight of the drug

Answer: B

40. The overall conclusion of Experiment 26 is that:

A. Convulsant agents have identical cellular mechanisms B. Anticonvulsant drugs exhibit differential protection depending on the specific neurotransmitter pathway targeted C. Plant extracts can never possess anticonvulsant activity D. Barbiturates and benzodiazepines act on unrelated receptors

Answer: B